

COLOSSEUM · SOLANA

Vitals

A patient is dying on a clock and you decide what happens next.
Anyone can play. Nobody can fake the replay.

VERIFIABLE CLINICAL REPLAY RUST · SOLANA AGPL-3.0 + COMMERCIAL

THE PROBLEM

Skill has no passport.

Since 2024, ECFMG certification — the gate to US residency — requires graduating from a school accredited by a WFME-recognised agency. Accreditation is granted to *institutions*, country by country, and most countries do not have it.

4,350+

medical schools in the World Directory

~23

countries with a WFME-recognised agency

130–147

countries whose graduates apply

Every one of those graduates ran hundreds of scored clinical encounters that would have proved what they can do. All of it sits in one institution's database, unverifiable by anyone outside it, and is thrown away at the border. **The evidence exists. The trust layer does not.**

NOT A MOCKUP

This runs. Here is the output.

One scenario — anaphylaxis, a 19-year-old, ten minutes from the first bite. Three players, three tapes, replayed against the shipped physiology automaton.

treated adrenaline IM · O ₂ · supine · fluids · admit	WinDischarge
stood up adrenaline IM · stood the patient up · O ₂ · fluids · admit	WinDischarge · harm
no adrenaline antihistamine · steroid · wait	DeathArrest

The middle row is the one that matters. **Same outcome as the first, different leaf**, because the harm is on the record. The anchor captures how you got there, not just whether you won. And nobody scripted the third — antihistamines are what people reach for when they haven't recognised what they're looking at, and the simulation kills her for it.

```
sce_hash 4d5ee49cc13dd5af1badf4d601010b03e70ded6d2e0ebaf339a26e7d1616a703
treated  65c56e52d9f718aef60cb4e7f712443fe779b10ffb3c3ac2e301464ebd275398
stood up  a6bec396869e7d889686bc42111f4a2b090aa16c4d93213262d24c858637f8ca
```

HOW IT PLAYS

The replay is the trophy, not the score.

XP and levels are the credential layer, and nobody has ever told a friend about a progress bar. What a run actually produces is a watchable ninety seconds that a stranger can re-derive — so the mechanics that matter are the ones determinism hands us for free.



GHOST · MOCKUP

`you · adrenaline 1:12` `ghost · 0:38`

Ghost racing. The tape replays exactly, so another player's run can run beside yours — and you watch the two patients diverge while you are still deciding. Trackmania's ghost car, in an emergency room.



RECORD · MOCKUP

`harm · stand/walk collapse` 50 saves · 12 harms

Harm is on the record. Fifty saves with twelve harms reads differently from thirty with none — and it keeps the fastest run and the cleanest run from being the same run.



DEATH CARD · MOCKUP

`DeathArrest · turned at 2:14``gold path → adrenaline IM at 0:40`

Failure is content. "She died because you reached for antihistamines" is a better clip than any win. A loss should be legible, teach in the same breath, and be worth showing someone.

The line that has to hold: the fun layer and the credential layer share data and must never share incentives. You rank on the **3R** board; a verifier stakes **\$VIGIL** to be allowed to say your run happened. Grinding a favourite episode buys a better *time* — the distinct-case gate makes sure it never buys a better *standing*.

THE CASE, AND THE MACHINE UNDER IT

The model makes it worth playing. The automaton makes it worth proving.

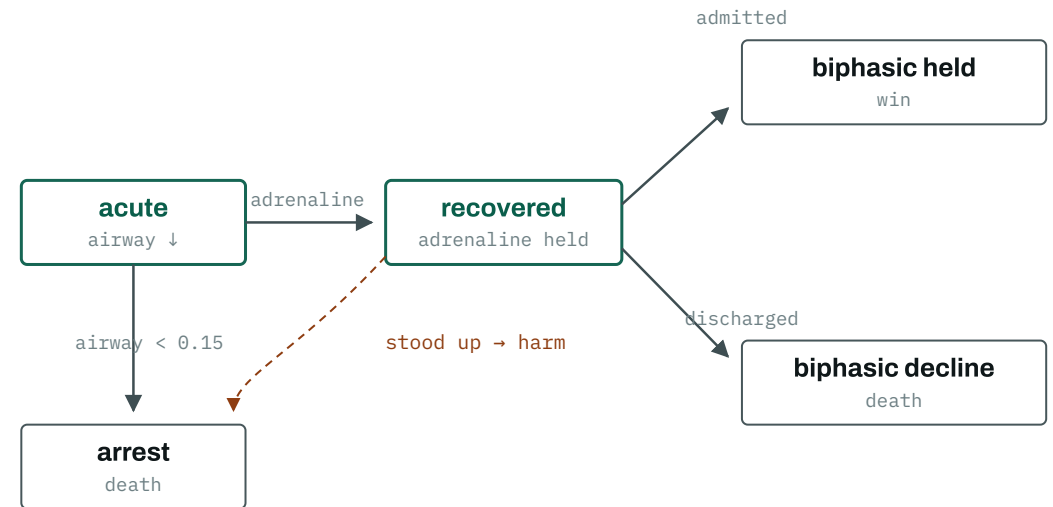
THE SITUATION

patient	19f · shellfish, ten minutes ago	HR 128
airway	stridor, voice going hoarse	SpO ₂ 91
shock	flushed, dizzy, about to faint	SBP 88

WHAT TREATING HER LOOKS LIKE

- **Adrenaline, intramuscular.** One drug decides this. Antihistamines and steroids are what people reach for when they haven't recognised it.
- **Oxygen, fluids, and lay her flat.** The ventricle is empty — *standing her up can kill her*.
- **Admit her.** Anaphylaxis relapses hours later. Sending her home is a decision the record has to remember.

THE SAME THING, AS A MACHINE



Every edge is a decision, every terminal is an ending. Nothing here is a score — it is what happened.

THE AGENT

Plays the patient — hesitates, hides the smoking history, is frightened. It decides **what you learn**. It is in the experience and never in the evidence.

THE AUTOMATON

Decides **what happens** — vitals, transitions, harm, ending. Exact, replayable, and the only thing the hash commits to.

AND THEREFORE THE TOKEN

Because the automaton is exact, **two honest verifiers cannot disagree**. That is what makes a fraud proof settle by arithmetic — and what \$VIGIL bonds.

THE MECHANISM

Every run reduces to one hash.

1 · COMMIT

Before play: `hash(scenario || player || nonce)` onchain. The chain knows how many attempts were *started*, so grinding for a good run is visible.

2 · PLAY

A deterministic automaton — vital dynamics, triggers, transitions, harm, terminal outcome. The tape is what you did and when.

3 · REDUCE

Discrete facts only: ordered beats, harm events, outcome. No float trajectory enters the hash, so nothing drifts across machines.

4 · ANCHOR

One compressed leaf. Anyone holding the tape and the scenario recomputes it — forever, without us.

```
leaf = sha256( sce_hash || tape || beats || harm_events || outcome )
```

There is no model anywhere in that path. An LLM plays the patient's dialogue in other modes and never touches the evidence. This is verifiable replay — what chess engines and speedrun leaderboards have done for decades — applied to a patient who is dying on a clock.

WHY THIS NEEDS A CHAIN

Watch the program refuse its own user.

Progression is not granted by a server. The program is handed the attempts, **recomputes the level itself**, and writes only what its own arithmetic agrees with. Live on a validator, today:

CLAIM: PROFICIENT

REJECTED — claim rejected: claimed Proficient, computed Competent
from 3 attempts (3 distinct, avg 7333bps)

CLAIM: COMPETENT

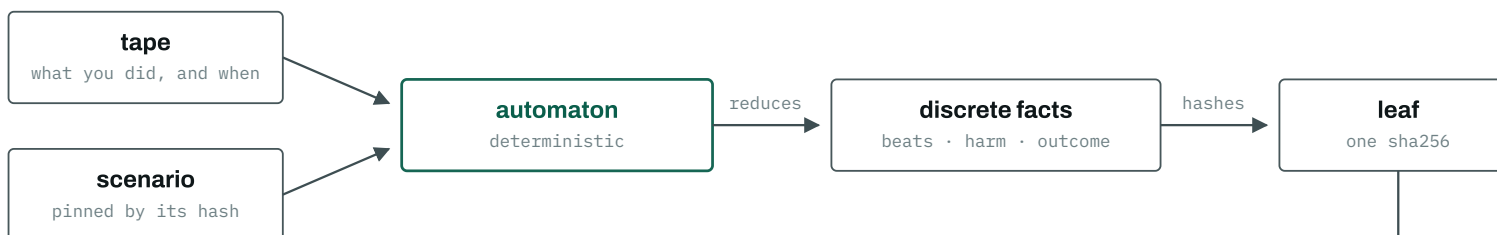
GRANTED — stored level Competent, distinct 3, xp 35. The level on chain is the one the program computed, never the one that was claimed.

No issuer key. No oracle. No authority that could be leaned on, bought, or subpoenaed. Take the chain away and you are back to trusting one company's database — which is precisely why digital credentials are worthless today.

ARCHITECTURE

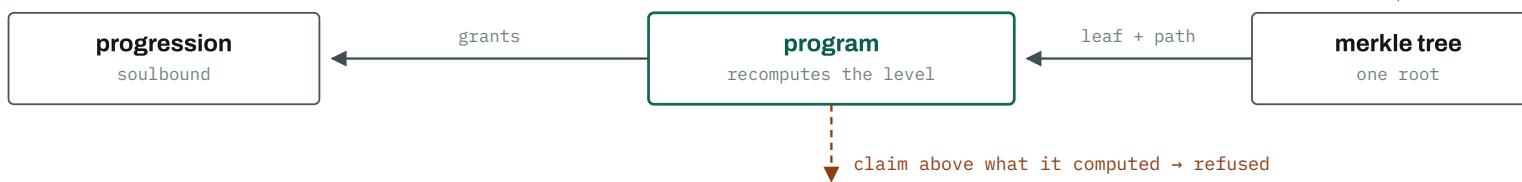
Two inputs in, one hash out, and nothing trusted in between.

OFF CHAIN



no model anywhere on this path — anyone holding both inputs recomputes the same leaf

ON CHAIN



The tape and the scenario are the only inputs. The chain never sees a run — only a leaf that has to prove against a root it built itself.

ONCHAIN ARCHITECTURE

Four primitives, none of them invented here.

LAYER	WHAT IT DOES	SOLANA PRIMITIVE
Scenario registry	Authors publish; content stays off-chain, only hash + price + royalty split onchain	Anchor program · Token-2022
Replay anchor	One compressed leaf per run — the high-volume write	Bubblegum state compression
Progression	Levels, skill trees, badges — minted permissionlessly, predicate recomputed onchain	Token-2022 <i>NonTransferable</i>
Credential	Accredited issuer attests an EPA competency to the player's wallet	Attestation Service

VOLUME

A single cohort year across the World Directory is hundreds of millions of runs. Compression puts that at **~\$110 per million**. Anchoring only the exams would be the cheap-chain compromise and would kill the thesis.

LATENCY

The skill tree advances in the same screen as the outcome. On a slow chain that becomes a pending spinner and stops feeling like a result.

ONBOARDING

Fee-payer relay. A player never sees a seed phrase, never holds SOL, never learns any of this is happening.

The token secures verification. It never represents competence.

Someone has to replay the tape and attest the outcome. Make that one party and you have rebuilt the database we are trying to escape. **\$VIGIL is what makes verification permissionless and expensive to lie in.** A vigil is kept beside the patient, never of the patient — which is exactly where this token sits relative to the credential.

THE LOOP

STAKE

Operators bond \$VIGIL to accept replay work. Bond size sets how much throughput they can take.

VERIFY

Replay is deterministic, so **two honest verifiers must agree**. Fees are paid by institutions and sponsors — in USDC, never by players.

DISPUTE → SLASH

Disagreement is provable: anyone posts a bond and re-runs the tape. The wrong verifier is slashed, the challenger is paid from it. Fraud proofs work here because the computation is exact.

SINKS · SOURCES

sink	verifier bond (locked while active)	supply out
sink	scenario listing + dispute bonds	supply out
sink	slashing — burn share	burned
source	verification fees → operators	earned
source	replay royalties → scenario authors	earned
note	authoring is cheap, so the author market is real	—

NEVER TOKENIZED

- **Credentials, badges, skill trees.** Token-2022 NonTransferable, no exceptions — a tradeable "Expert in Cardiology" is credential fraud with extra steps.
- **Access to play.** No token gates a student out of practice, ever. Players never hold \$VIGIL and never see it — the arena has its own name, **3R**, so the two layers do not even share a word.

Allocation is a proposal, not a commitment, and no token ships inside the hackathon window: the verification market only has to exist once verification is worth attacking. Distribution weighted to operators and scenario authors — the two parties whose work the network actually consumes.

MARKET AND PRECEDENT

This shape already won this hackathon once.

FRONTIER 2026 · GRAND CHAMPION

CrowdBrain — "a vertically-integrated robotics DePIN built to **train users in simulation, qualify them through QA, and route the best operators to real robots.**"

Train in simulation → verify competence → route the qualified to real work. That is Vitals, for robot operators instead of clinicians. We think the pattern generalises; the largest crypto hackathon ever run agreed three months ago, out of 2,857 submissions from 150 countries.

HOW EPISODE SIX GETS MADE — SKALD

A producer console. Type a premise and a number of episodes and an AI showrunner drafts the series bible and scaffolds every storyboard. Then each shot runs a loop: write the prompt, generate, **a crew fails the shot before the video model ever sees it** — a medical advisor on the procedure, a continuity supervisor on the face — approve, and every generation is kept as a take you can roll back to, like a film set. A budget gate blocks spending past its ceiling.

Same principle as the rest of this: something that must be right, checked automatically. **Skald verifies the content. The chain verifies the play.**

WHY MEDICINE FIRST

The hardest instance: highest stakes, heaviest regulation, most protective of data. Solve it here and it generalises downward to every other skill credential.

WHO PAYS

Institutions on ordinary invoices, sponsors funding outcome-linked scholarships, and platform fees on the scenario market. Players pay nothing and hold nothing.

HEAD START

A clinical simulator already in production — 424 authored scenarios, deterministic scoring, a hash-chained audit log, real students.

WHERE IT STANDS, AND WHAT'S NEXT

Two halves built. One week to join them.

WORKING NOW

- **Replay** → **leaf**. EP1 runs against the shipped automaton; three tapes, three distinct leaves, reproducible across runs.
- **Program** → **refusal**. Native Solana program live on a validator, recomputing levels and rejecting claims that don't survive its own arithmetic.
- **Rust end to end** — one scoring implementation compiled into the game, the verifier, and the program.

THE REMAINING FOUR WEEKS

- **W1** — join the halves: `claim_progress` takes merkle proofs against anchored leaves instead of attempts handed to it.
- **W2** — compressed anchors, scenario registry, public verify page.
- **W3** — soulbound progression, badge-gated scenarios, sponsor escrow paying out on a provable level.
- **W4** — mainnet-beta, demo video, open-source release.

Cut on purpose

ZK selective disclosure · verifier DePIN with live staking · token launch · mobile · new clinical content. Each is a real v2 line and none survives four weeks. The credential layer stays a stub until an accrediting institution is standing next to it — progression doesn't need one, which is why progression is the demo.

The 60-second version: you play a patient who is dying, the chain keeps a replay of what you did, and anyone can re-derive it without asking us.